

Department of Computer Science



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Science

Evaluating the Impact of Software-Based Motor Accessibility in Digital Games

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STATEMENT OF ETHICS

This project involved voluntary participant playtesting and feedback. Participants were informed of the study purpose before taking part and were able to withdraw at any time. No sensitive personal or medical information was collected.

All collected data was anonymised and used only for the purposes of this dissertation. Gameplay metrics, questionnaire responses, and interview comments are presented without identifying participants. The project also avoids making direct claims about disabled players lived experiences, as participants were not recruited based on motor impairment.

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Executive Summary

This project investigates how software-based motor-accessibility features affect gameplay performance, usability, and player experience in a digital game prototype. The work is motivated by the fact that many mainstream games rely on rapid, repeated, sustained, precise, or simultaneous inputs, which can create barriers for players with motor impairments. Although accessibility guidance recommends features such as simplified controls, input reduction, automation, and one-handed play, there is limited practical evidence showing how these features affect gameplay when implemented in a working game system.

To explore this, a 2D top-down action game prototype was designed and developed in Unity. The game required players to move, aim, shoot, reload, dodge, avoid enemies, and gain score by eliminating enemies. Two control schemes were implemented. The standard scheme used conventional keyboard and mouse controls, including WASD movement, mouse aiming, repeated clicking to shoot, manual reloading, and double-tap dodging. The accessible scheme kept the same gameplay objectives but reduced motor demands through click-to-move movement, automated shooting, automated reloading, and single-click dodging.

The prototype was evaluated using a mixed-methods, within-subjects playtest with eight participants. Each participant played both versions of the game. Data was collected through gameplay metrics, input tracking, questionnaires, observation, and post-playtest interview questions. The evaluation focused on three areas: gameplay performance, usability, and player experience.

The accessible control scheme substantially reduced input demand. Total inputs fell by approximately 85%, while simultaneous, sustained, and repeated inputs were almost entirely removed. Participants also reported lower physical effort, lower fatigue, and reduced control complexity in the accessible version.

Gameplay performance improved overall. Every participant achieved a higher score with the accessible controls. Average score increased by 16.69%, enemy eliminations increased by 11.27%, deaths decreased by 34.62%, and dodges increased by 35.16%. Statistical testing showed significant improvements for score, deaths, eliminations, dodges, and input demand metrics. However, the results

Executive Summary

also showed that the accessible scheme changed how participants played. By making shooting and dodging easier, it encouraged a more offensive playstyle, showing that accessibility features can influence gameplay strategy as well as input effort.

Usability also improved. Participants generally found the accessible controls easier to learn, simpler to use, and less demanding to manage during intense gameplay. However, some trade-offs were identified. Because movement and aiming both relied on the mouse, some participants felt they had less precise control compared with the standard scheme.

Player experience showed a similar balance. The accessible version reduced frustration and made the game more comfortable. However, this made the experience less intense, less challenging, and less directly controlled for some participants. This suggests that accessibility features can change the feel, challenge, and agency of play.

Overall, the project found that software-based motor-accessibility features can reduce input demand, improve usability, and support better performance. However, they must be designed carefully because they can also affect strategy, challenge, control, and player experience. The main contribution of the project is showing that accessibility should be treated as part of core game design, not as an optional addition. Since the study used a general participant sample rather than players with motor impairments, the findings are exploratory and should be validated through future testing with the intended user group.

Ethical and professional considerations were addressed through voluntary participation, anonymised data collection, and careful interpretation of findings. No sensitive medical information was collected, and the study avoids making direct claims about disabled players lived experiences. No significant legal or commercial issues were identified beyond responsible data handling and accessibility-aware design practice.

1 Introduction

Digital games represent one of the largest segments of the global entertainment industry, with projections estimating that the global gaming market will exceed three billion users by 2030 [1]. As games become increasingly widespread, ensuring they can be accessed and enjoyed by diverse player populations has become an important consideration within game design. Despite this, accessibility barriers remain persistent. Survey evidence indicates that 66% of players with disabilities have encountered barriers while playing games [2]. Although research and industry guidelines increasingly recommend configurable accessibility features, these are not consistently implemented across commercial titles. Existing research also provides limited comparative evidence showing how accessibility features affect performance, usability, and player experience in practical gameplay contexts.

This project aims to design and develop a game prototype incorporating software-based motor-accessibility features informed by academic research and industry guidance. The prototype is evaluated to examine the design impact of these features on performance, usability, and player experience.

1. Synthesise current research and industry guidance relating to motor accessibility in digital games.
2. Design and develop a game prototype incorporating representative gameplay mechanics that elicit common motor barriers.
3. Implement selected software-based accessibility features within the prototype that reduce input demand and complexity.
4. Evaluate the prototype using objective and subjective measures to assess the impact of the accessible control scheme.

Through this approach, the project provides a comparative assessment of accessible design within a controlled gameplay context. The findings contribute practical insights into how accessibility adaptations affect performance, input demand, usability, and player experience, supporting a more holistic understanding of the benefits and trade-offs involved in software-based motor accessibility.

2 Literature Review

2.1 Background

Digital games play an important role in entertainment, identity formation, and social connection for players worldwide [3]. However, accessibility barriers prevent many individuals from fully participating in this growth. While estimates vary, industry research suggests that around 20% of gamers identify as having some form of disability [4], [5], representing hundreds of millions of current and potential players. Motor impairments are a particularly significant source of exclusion because games frequently rely on rapid, precise, or simultaneous physical inputs. For players with reduced motor control, these input demands can make interaction physically demanding or inaccessible. As a result, inaccessible design may restrict participation in certain genres, multiplayer environments, or gaming entirely [6].

2.2 Motor Impairments and Gameplay Barriers

Research consistently identifies motor demands as a significant source of exclusion within mainstream digital games. Early survey work on game accessibility demonstrated that many disabled players are unable to perform the inputs required by common gameplay mechanics [8], [9]. These input demands include sustained button holds, repeated button presses, time-sensitive reactions, analogue stick precision, and multi-button coordination across both hands. Motor impairments may affect strength, endurance, coordination, or fine motor control, meaning that even mechanics considered standard within mainstream design can become physically demanding. Empirical and qualitative studies report fatigue, strain, frustration, and reduced progression as common consequences of sustained motor effort [10], [11].

2.3 Accessibility Solutions

Both academic literature and industry guidelines emphasise the integration of software-based accessibility features directly within games [10], [7], [12], [13]. While adaptive hardware can also mitigate some barriers, cost, configuration demands, and game compatibility issues mean that hardware alone cannot ensure consistent accessibility across titles [2], [8], [11]. Software can modify interaction demands at the design level and can be standardised across releases. Several recurring features are consistently recommended. Control remapping enables players to reposition difficult inputs to accessible configurations and is frequently cited as a fundamental accessibility requirement [10], [7]. Toggle alternatives to sustained holds reduce physical strain by eliminating prolonged button presses, which may otherwise cause fatigue or discomfort [10], [11], [9]. Simplifying input

complexity such as reducing rapid presses and simultaneous actions can further decrease motor demand [10], [7]. Additional features such as one-handed play support and aim assistance reduce coordination and precision requirements [12].

Comparing a range of accessibility guidelines [7], [12], [13] shows that many methods for reducing motor barriers are already well established. However, their implementation remains inconsistent across mainstream games, signifying a weakness in accessibility guidance and routine development practice. Accessibility features are often treated as optional settings rather than core interaction design principles, resulting in solutions that are limited in scope, unevenly implemented, or absent entirely [10], [14]. This inconsistency emphasises the need to examine how these features function when integrated directly into gameplay systems rather than considered as supplementary.

2.4 The Wider Impact of Accessibility in Games

Accessibility in games should be understood as more than the removal of mechanical input barriers. While reducing physical demand is essential, games are also interactive experiences shaped by challenge, agency, competence, engagement, and perceived control. This is particularly important in action-oriented games, where timing, aiming, movement, and input execution often contribute to the player's sense of skill and accomplishment. Player experience research supports the need to evaluate these broader dimensions. The Game Experience Questionnaire identifies components such as immersion, flow, competence, affect, tension, and challenge as important aspects of gameplay experience [31]. These concepts are relevant to accessibility because features that simplify or automate interaction may reduce physical effort while also changing how players experience control, skill, and challenge. The MDA framework supports this wider view by defining mechanics as the actions and control mechanisms available to the player, which give rise to gameplay dynamics and ultimately shape player experience [33]. As a result, accessibility features that alter how actions are performed may affect whether players can interact with a game and how they engage with its mechanics, strategies, and playstyles.

This creates a design tension. Accessibility features should reduce unnecessary barriers, but they should also preserve meaningful interaction where possible. They may make gameplay more usable while altering the player's overall experience or the strategy they adopt. Evaluating accessibility features therefore requires more than measuring whether players perform better. It also requires examining how those features affect usability, engagement, challenge, and perceived control.

This project adopts this broader view by evaluating software-based motor-accessibility features through performance metrics, input demand data, usability responses, and player-experience feedback. This allows the study to examine whether the accessible control scheme reduces interaction demands and whether it introduces trade-offs in the experience of play.

2.5 Limitations of Existing Research

Despite growing awareness of game accessibility, key research areas remain limited in the literature, particularly regarding motor impairments. Although accessibility research in games has expanded in recent years, most scholarly output focuses on visual impairments, while motor accessibility receives comparatively little attention. Given the prevalence and impact of motor barriers, this represents a significant oversight in the field [14]. While studies frequently identify gameplay barriers for motor-impaired players, there is limited experimental evidence testing how software-based accessibility features influence gameplay outcomes. Comparative analysis between standard and modified control schemes is rare [8], [14], [11]. As a result, the effectiveness of accessibility features is often assumed rather than empirically evaluated. This creates a limitation between identifying recommended accessibility practices and understanding how those practices affect gameplay when implemented within a working game system. Reviews of accessibility research highlight that evaluation approaches are highly variable, with few studies combining measures of usability, performance, and player experience into a unified framework [14], [15]. This inconsistency hinders efforts to systematically assess the effectiveness of motor accessibility inclusivity techniques.

A final challenge concerns participant access and evaluation scope. Testing with motor-impaired players is essential for validating the real-world accessibility of a system, as the intended user group is best placed to assess whether features meet their needs and preferences. Accessibility research emphasises the importance of involving representative users, and disability-centred design approaches caution against making claims about disabled users without their direct involvement [34]. However, controlled evaluations with a general participant sample can still provide useful evidence about the design impact of accessibility features [35]. Such findings should not be treated as conclusive evidence of accessibility effectiveness for motor-impaired players. Therefore, this study does not claim to represent the lived experience or accessibility needs of motor-impaired players; instead, it uses a comparative evaluation to examine how accessibility-informed features affect interaction and gameplay outcomes under controlled conditions. This can identify design trade-offs and inform future testing with the intended user group. Together, these limitations

show the need for controlled empirical evaluations of software-based motor-accessibility features in mainstream gameplay contexts. This project responds to that limitation by implementing a prototype with standard and accessible control schemes and evaluating their impact on performance, input demand, usability, and player experience. The study therefore investigates the design impact and potential accessibility value of these features, while recognising that further work with motor-impaired players is required to validate real-world accessibility effectiveness.

2.6 Synthesis and Research Direction

Synthesising the literature in accordance with the first project objective, several conclusions can be drawn. Motor impairments create significant barriers in mainstream gameplay, particularly where games require rapid, precise, sustained, or simultaneous inputs. Software-based accessibility features are widely recommended to reduce these demands. However, the literature provides limited comparative evidence showing how these features affect gameplay when implemented within a working game system. There is a need to evaluate whether accessibility features reduce input demand, and their wider impact, exploring how they influence performance, usability, and experience. The evaluation uses a general participant sample and does not claim to directly represent the lived experience of motor-impaired players. Instead, it examines whether features informed by motor-accessibility guidance reduce interaction demands and how they affect gameplay outcomes and player experience. The findings are interpreted as exploratory evidence of potential accessibility value.

The following research questions guide the projects fourth objective to empirically evaluate the prototype, assessing the impact of accessibility features.

In a controlled game prototype:

RQ1. What impact do software-based motor-accessibility features have on gameplay performance?

RQ2. How do software-based motor-accessibility features affect usability?

RQ3. What impact do software-based motor-accessibility features have on player experience?

3 Methodology

3.1 Research Approach

This project adopts an applied software engineering approach situated within the Human-Computer Interaction (HCI) paradigm. Findings from accessibility research are translated into a functional game prototype and empirically evaluated. Within this framework, the prototype functions as an experimental artefact, enabling the systematic investigation of how accessibility features influence performance, usability, and player experience. To support evaluation, the prototype is designed with two fixed conditions: a standard control scheme and an accessible control scheme incorporating reduced motor demands.

3.2 Methodology Selection

A structured development approach based on the Waterfall model [17] was used to guide implementation of the prototype. This approach was selected due to the fixed scope, predefined requirements derived from literature, and the need to produce a stable and consistent system prior to evaluation. In contrast to iterative methodologies such as Agile, which emphasise continuous refinement through user feedback, this project required a controlled and unchanging system to support valid comparative analysis. Introducing iterative changes during development could have altered gameplay mechanics or input behaviour between testing phases, introducing confounding variables and reducing the reliability of results. By implementing accessibility features in a single, predefined development cycle, the system maintains consistency across all participants and conditions, ensuring that observed differences in performance, usability, and player experience can be attributed directly to the control schemes rather than ongoing design changes. This approach also supports clear traceability between research findings, system design, and evaluation outcomes [18], [19].

Accessibility features were therefore identified in advance from academic research and industry guidelines and implemented systematically. This ensures a direct relationship between identified gameplay barriers, implemented solutions, and evaluation measures, strengthening the validity of the experimental design.

3.3 Requirements Derivation

3.3.1 Gameplay Requirements

The prototype must include a representative gameplay structure capable of generating measurable motor demands as required by the second project objective. Game studies literature conceptualises digital games as rule-based interactive systems defined by player

input, system response, and measurable outcomes [20], [21]. Within this framework, a simplified prototype can meaningfully represent broader gameplay dynamics if core interaction loops are preserved. To maintain realism while ensuring manageability, the prototype incorporates a focused and engaging gameplay loop that encompasses the gameplay requirements seen in Appendix A, Table A.1. FR-03 to FR-07 define the core player actions, including movement, aiming, shooting, reloading, and dodging. These were selected to ensure that the system incorporates interaction patterns representative of mainstream games, while also introducing varied motor demands such as sustained, simultaneous, and repeated inputs. These actions are later adapted across both control schemes to enable direct comparison of how differing input designs influence motor effort. FR-08 to FR-11 define enemy behaviour and interaction, providing the primary source of gameplay challenge. Enemies continuously pressure the player, requiring frequent use of the defined actions, thereby eliciting the motor demands and potential barriers identified in the literature. This ensures that player interaction is both necessary and sustained throughout gameplay, rather than optional. FR-12 defines a scoring system, where players gain and lose points based on enemy eliminations and collisions. This provides a measurable representation of gameplay performance, enabling comparison between control schemes in terms of effectiveness and success. Together, these requirements establish a controlled gameplay loop that generates consistent interaction demands while producing quantifiable outcomes for evaluation.

3.3.2 Accessibility Requirements

Based on the evaluation approach, both a standard and an accessible version of the game are required. As discussed in Section 2.2 the literature identifies several common motor barriers within mainstream gameplay, informing the identification of functional accessibility requirements shown in Table A.2. Mitigation of these barriers were subsequently converted into system requirements FR-13 to FR-17, the design and implementation of which will achieve the project's third objective to develop selected software-based accessibility features within the prototype.

3.4 Evaluation Methodology

3.4.1 Experimental Design

As guided by the project's fourth objective, a comprehensive evaluation of the prototype is required. A mixed-methods approach, combining quantitative performance metrics with qualitative user feedback, was adopted to evaluate the impact of accessibility features on gameplay. The integration of both quantitative and qualitative data provides a more comprehensive understanding of interaction quality,

addressing the limitations associated with relying on a single form of data collection [22]. In the context of game evaluation, playtesting is widely used to identify usability issues and interaction barriers. Playtesting can be used as a process for identifying aspects of a game that prevent players from effectively understanding or using its mechanics [23]. Observing real users interacting with a system is recognised as one of the most effective methods for identifying usability issues [24]. It allows problems to be captured as they happen, rather than only relying on what participants report. This is appropriate because the evaluation focuses on how players use the two control schemes during gameplay, including where they struggle, how much effort is required, and how the controls affect their experience.

The evaluation is structured using a within-subjects design, where each participant experiences both the standard and accessible versions of the game. This approach is commonly used in HCI research when comparing interface variations, as it allows participants to act as their own control, reducing the influence of individual differences such as gaming experience or motor ability [25]. It is particularly suitable for studies with smaller sample sizes, as it increases statistical power and enables direct comparison between conditions, aligning with this project's objectives. A counterbalancing strategy was considered, in which half the participants would play the first version first, and the other would play the second version first. This would help mitigate order and fatigue effects [25]. However, counterbalancing was not implemented in this study. Instead, all participants experienced the conditions in the same fixed order to ensure a consistent and structured playtesting procedure. This trade-off prioritised procedural consistency over experimental control, reducing variability introduced by differing session flows. This decision may introduce order effects, such as learning or fatigue influencing performance and experience in the second condition. These biases are acknowledged as a limitation and will be considered when interpreting the results. Overall, this approach directly addresses the limitation identified in Section 2.5 regarding the limited use of comparative evaluations in motor-accessibility research.

3.4.2 Playtesting

The evaluation is conducted through a structured playtest for both versions. This approach was selected over uncontrolled observation to ensure consistency across participants and minimise variability in interaction conditions. This is particularly important given the within-subjects design, where comparability between conditions is critical. During gameplay, participants are encouraged to follow a think-aloud protocol, in which they verbalise their thoughts while interacting with the system. This method was selected for its ability to reveal the

underlying causes of user errors and interaction difficulties, providing insight beyond what can be inferred from performance metrics alone [26]. Gameplay sessions are screen-recorded to support detailed post-session analysis (found in Appendix G, Figure G.4). This reduces reliance on real-time notetaking and ensures that transient behaviours and interaction issues are not overlooked, aligning with recommended usability evaluation practices [24], [27].

3.4.3 Participants

Given the exploratory scope and practical constraints of this study, participants were recruited using convenience sampling, with the aim of capturing a range of gaming experience levels. This was important because experience may influence how easily participants adapt to each control scheme, introducing important considerations for evaluation. Participants were not required to have diagnosed motor impairments. Acknowledging the limitations discussed in Section 2.5, the study uses a general participant sample to examine how accessibility features impact the gameplay aspects defined by the research questions. Further testing with motor-impaired players would be required to validate whether the features meet the needs of the intended user group.

Prior usability research suggests that the required number of participants depends on the purpose of the study. When the aim is to produce statistically significant quantitative results, at least 20 participants are recommended, with larger samples required for tighter confidence intervals [37]. However, qualitative usability studies can identify many major usability issues with a much smaller sample, commonly around five participants [28], [29]. Other studies challenge this claim, arguing that such justifications rely on simplifying assumptions and that additional variables can significantly influence the number of participants required [36]. In this project, eight participants were considered appropriate for the scope and purpose of the evaluation. The study did not aim to produce statistically conclusive or generalisable quantitative results. Instead, the quantitative data was used to support insight into patterns of performance, input demand, and user behaviour, alongside questionnaire responses, observation, and qualitative feedback. This decision reflects a balance between established usability research practices and the need to account for potential variability, while remaining feasible within the scope of the project, particularly because each session required structured observation, screen recording, condition consistency, and post-session questioning.

3.4.4 Procedure

Each playtesting session follows a structured and consistent process. Participants are briefed on the study, provide informed consent, and then complete two gameplay sessions (standard and accessible conditions). Each session begins with a one-minute tutorial that introduces the controls and gameplay mechanics. Participants then move onto the main game, playing for three minutes, with a short break provided between sessions to reduce the limitation of fatigue effects discussed earlier. Immediately after each session, participants complete a questionnaire concerning the version. Conducting questionnaires after each condition reduces memory bias and enables more accurate reporting of user experience. After both sessions participants are asked open-ended interview questions. This allows for deeper qualitative insights into perceived differences between conditions.

3.4.5 Data Collection

To ensure a comprehensive evaluation of motor accessibility, data collection combines objective performance metrics with subjective and comparative user feedback. This approach enables the assessment of both measurable interaction outcomes and perceived player experience, providing a holistic understanding of how accessibility features influence gameplay. The performance metrics recorded are deaths, score, enemy collisions, enemies eliminated, shots fired, accuracy, and dodges performed. These metrics represent overall gameplay performance and effectiveness. Within the context of this study, these metrics provide a baseline for comparing how accessibility features influence outcomes of play, directly addressing RQ1. To ensure that these measures are grounded in system design, Table 3.1 identifies the functional requirements that contribute to each performance metric. This establishes a clear relationship between gameplay mechanics and measurable outcomes, allowing performance data to be interpreted in terms of the specific interactions that produce it.

Performance Metric	Contributing Requirement
Deaths	FR-11
Score	FR-10, FR-11
Enemy Collisions	FR-11
Enemy Eliminations	FR-10
Shots Fired	FR-05
Accuracy	FR-04
Dodges Performed	FR-07

Table 3.1: Requirements Contributing to Performance Metrics

Performance metrics alone are insufficient to evaluate accessibility. A player may achieve similar performance in both versions while experiencing significantly different input demand. For this reason, performance data must be interpreted alongside interaction-based measures to provide evidence supporting the effectiveness of the accessible scheme. Table 3.2 maps these metrics to the functional requirements that generate them, establishing a direct link between player input behaviour and system design. The interaction metrics recorded are total number of inputs, simultaneous inputs, sustained inputs, and repeated inputs.

Input Metric	Contributing Requirement
Total Number of Inputs	FR-03 to FR-07
Simultaneous Inputs	FR-03 to FR-07
Sustained Inputs	FR-03
Repeated Inputs	FR-05, FR-07

Table 3.2: Requirements Contributing to Input Metrics

By measuring how players interact with the system, these metrics provide insight into efficiency, effort, and usability. Combining these metric groups makes the evaluation more meaningful. It allows analysis of the relationship between performance outcomes and interaction effort. For example, if players achieve similar or improved performance with fewer inputs or reduced input complexity, this suggests that accessibility features have successfully reduced motor demands without negatively impacting gameplay. This approach directly reflects the study's research objective and aligns with earlier literature, which emphasises that accessibility should reduce interaction difficulty and physical demand, rather than simply maintain performance outcomes [10], [7].

Subjective data is collected using a post-session questionnaire. This assesses perceived difficulty of controls, usability, and overall enjoyment and engagement, contributing to the projects research questions. Questions were selected and adapted from a range of validated questionnaires, including the Game Experience Questionnaire (GEQ) [31] for RQ3, the System Usability Scale (SUS) [32] for RQ2, and challenge-based measures from the CORGIS framework [30], supporting RQ1 and RQ2 by providing additional insight into the input demands of each version. These were chosen for their ability to capture key dimensions relevant to this study. The full list of questions can be found in Appendix B, Figure B.1.

Final open-ended interview questions (Figure B.2) ask participants to directly compare the two gameplay conditions. Overall experience and perceived differences in the control schemes are captured first to provide a generalised context. Specific insights into RQ2 and RQ3 are

then explored in depth to allow participants to elaborate on if usability was improved, and if player experience was maintained. This structured approach ensures that initial impressions are contextualised before more focused discussion, enabling richer qualitative data that supports interpretation of the quantitative findings.

Comparative questioning is particularly important in this study because the evaluation focuses on how the accessible control scheme differs from the standard version. While gameplay metrics show what changed, they do not fully explain how participants experienced those differences. Asking participants to compare both versions helps identify perceived changes in effort, control, usability, challenge, and enjoyment, directly supporting the project's research questions.

3.5 Ethical Considerations

As the study involves gameplay tasks, there is a potential for minor fatigue or frustration, particularly in the standard control condition where higher motor demands are present. To mitigate this, session durations were kept short, and breaks were provided between conditions. Additionally, as the study relates to accessibility and motor impairments, care was taken to ensure that the design and evaluation of the prototype did not misrepresent or oversimplify accessibility needs. Participants were not required to have motor impairments; the sample was not intended to replace evaluation with motor-impaired players. Instead, it was used to explore how accessibility design affected performance, usability, and player experience under controlled conditions. The accessibility features implemented were derived from established research and industry guidance, rather than assumptions, to ensure that identified barriers and solutions were represented as accurately as possible.

4 Design

4.1 Design Approach

The design is driven by the requirements derived in Section 3.3, ensuring traceability between them and system functionality. Additionally, the system is designed to support a controlled, within-subjects evaluation approach (Section 3.4.1), requiring consistent gameplay conditions across both control schemes. A full mapping from requirement to design is provided in Appendix C, outlining how each gameplay and accessibility requirement was translated into a corresponding design decision. This chapter summarises the key design choices most relevant to the evaluation.

The prototype was developed using the Unity game engine. Unity was selected due to its suitability for efficient and simple development of 2D interactive systems, and its support for input handling and configurable control schemes. Additionally, prior experience with Unity reduced development risk and enabled greater focus on the design and evaluation of accessibility features.

4.2 Design Overview

The core design consists of continuous player movement, enemy avoidance, and combat. The player moves within the 2D top-down environment. This perspective was selected to provide a simplified and controlled representation of gameplay interaction. It removes the need for camera control, helping isolate motor interaction as the primary variable. The environment is open and unobstructed, with simplified visuals and no audio, so usability issues can be attributed mainly to input demands. Overall, these choices support NFR-01 by minimising environmental, visual, and audible complexity and ensuring that gameplay focuses on interaction mechanics.

The player aims using directional input and damages enemies by shooting projectiles. A reload mechanic introduces resource management, requiring the player to monitor ammunition during combat, while a dodge ability provides a defensive option for rapid repositioning. These mechanics were included to create a simple but complete action-game loop, where players must balance movement, attacking, avoidance, and reactive decision-making.

Enemies repeatedly spawn in the environment, continuously move towards the player, applying constant pressure to elicit motor demands. Enemy behaviour was deliberately simplified to maintain a controlled interaction environment. More complex behaviours, such as ranged attacks, were excluded to avoid introducing confounding variables that could influence player performance independently of

input demands. This ensures that differences observed between control schemes can be attributed primarily to variations in motor interaction rather than changes in gameplay complexity.

Score increases by 100 through enemy eliminations, which also replenishes health, and decreases by 50 through collisions, providing a continuous indicator of player performance. A death system is also included, where the player loses all health upon repeated collisions with enemies. Rather than ending the session, the player is immediately respawned, and the death count is incremented. This design ensures that all participants experience a consistent gameplay duration. The combination of score and deaths encourages players to balance offensive and defensive behaviour, reinforcing the core gameplay loop of movement, aiming, and combat.

The design of the gameplay loop requires balancing experimental control with the need to produce a meaningful player experience. While a simplified system improves the ability to isolate input as the primary variable, excessive simplification risks limiting engagement, challenge, and interaction depth. This presents a particular issue for this study, as player experience forms a key part of the evaluation. To address this, a phase-based difficulty system was introduced. Rather than adding new mechanics, which could introduce confounding variables, difficulty is increased through adjustments to enemy count, speed, health, and spawn rate. This allows gameplay to become progressively more demanding while preserving the same core interaction loop. As a result, the system maintains experimental consistency while providing sufficient variation for differences in control schemes to emerge under increasing pressure. This loop was selected to generate continuous motor interaction while maintaining simplicity, ensuring that input demands are consistently elicited throughout gameplay.

4.3 Control Schemes

To support the evaluation methodology outlined in Section 3.4, the system implements two control schemes: a standard configuration and an accessible alternative (FR-13). Both schemes provide identical gameplay functionality while differing in input complexity and interaction demands, ensuring compliance with NFR-02.

4.3.1 Standard Control Scheme

Controls were selected that create varied motor demands aligning with barriers explored in Section 2.2, making them suitable for comparison. Player movement is controlled using the W (Up), A (Left), S (Down), and D (Right) keys, requiring sustained and simultaneous input to keep moving. Aiming is performed using the mouse, eliciting precision

requirements. Shooting is executed by clicking the left mouse button. Reloading is assigned to the R key, and dodging is performed by quickly tapping the spacebar twice, prompting further input demands that require repetition, timing, and reaction. This configuration requires simultaneous coordination between keyboard and mouse inputs, as well as sustained directional input and repeated button presses.

4.3.2 Accessible Control Scheme

Research synthesised in Section 2.3 indicates that industry guidance proposes several specific mitigation strategies to motor barriers. This section outlines how the accessibility requirements derived in Section 3.3.2 inform the accessible control scheme design. This is achieved by directly modifying input behaviour rather than altering gameplay mechanics, reinforcing compliance with NFR-02 by maintaining consistency between gameplay conditions.

Movement is controlled using the left mouse button; clicking moves the player in the direction of the mouse, allowing navigation through a single input interaction. Click-to-move is identified in accessibility guidance as a useful alternative to conventional movement controls for players with motor access needs [38]. Shooting is automated when the mouse is close to an enemy. Dodging is mapped to a single right mouse button click. Reloading is also automated when the player's weapon has 0 ammo left. These modifications are designed to reduce motor effort while preserving gameplay functionality, supporting NFR-03. Maintaining consistent responsiveness across both control schemes (NFR-04) ensures that changes in performance, player experience, and usability are solely due to motor demands. table 4.1 shows the control scheme layouts, the actions they correspond to, and the barriers elicited and consequently mitigated. This shows how each core gameplay action was preserved while its input requirements were modified to reduce barriers such as sustained input, simultaneous coordination, precision, repetition, and timed actions.

ID	Action	Standard	Barrier	Accessible
FR-03	Move	W, A, S, D	Sustained, Simultaneous	Single Left Mouse Click
FR-04	Aiming	Mouse.	Precision	Hitbox Assisted
FR-05	Shooting	Clicking Left Mouse.	Repetition	Automated
FR-06	Reloading	R	Timed	Automated
FR-07	Dodging	Double-tap Space	Repetition	Single Right Mouse Click

Table 4.1: Control Scheme Layouts

5 Implementation

5.1 Implementation Overview

The implementation translates the design defined in Section 4 into a functional prototype that supports controlled experimental evaluation. While the design specifies intended system behaviour, the implementation focuses on how these behaviours are realised within the constraints of the Unity engine, while maintaining consistency, modularity, and experimental validity.

A key requirement of the implementation was to ensure that both control schemes operate over identical gameplay systems. This constraint is critical to the evaluation methodology outlined in Section 3.4 as any variation in underlying mechanics would introduce confounding variables. As a result, the system was structured around a clear separation between input handling, gameplay logic, and metrics collection. This allows input behaviour to vary independently, while all other systems remain unchanged. Unity's component-based architecture supports this separation by allowing behaviour to be distributed across scripts. Gameplay entities such as the player and enemies are constructed from collections of components, enabling modularity and reuse. This structure also simplifies debugging and iteration, as individual systems can be modified without affecting the overall behaviour of the prototype.

5.2 Input System

5.2.1 Input Abstraction

To ensure consistency across control schemes, input handling was abstracted using a common interface. The `IInput` interface (Appendix F, Figure F.1) defines the core player actions identified in requirements FR-03 to FR-07, including movement, aiming, shooting, reloading, and dodging. This abstraction ensures that gameplay systems remain independent of specific input implementations, allowing control schemes to vary without modifying gameplay logic. This abstraction maintains experimental control, ensuring differences in implementation between control schemes can't introduce unintended variations in behaviour beyond input complexity.

5.2.2 Control Scheme Integration

Unity's Input System was used to define separate action maps, representing the standard and accessible control schemes defined in Section 4.3. The Unity setup for these can be found in Appendix E. These maps encapsulate the bindings and interaction logic required for each configuration, allowing them to be switched dynamically at runtime. A key implementation challenge was ensuring that both

Implementation

control schemes produced equivalent gameplay outcomes despite differing input structures.

While a one-handed accessible control scheme is recommended in accessible design [38] and implemented in this prototype, several technical constraints are introduced. Mapping multiple actions to a single input device required careful handling to prevent conflicting behaviour. For example, both movement and shooting depend on mouse input, creating potential ambiguity in user intent. Automated shooting was achieved with a hitbox-based detection system, which triggers shooting only when the cursor overlaps with an enemy-specific target area (see Appendix G, Figure G.1). This provided more predictable and intentional behaviour.

This implementation introduced an important trade-off. Automating shooting and reloading (Appendix F, Figures F.2 and F.3) reduces physical input demand but also reduces the player's direct control over combat timing. Similarly, click-to-move reduced sustained keyboard input, but made movement dependent on the same mouse behaviour used for aiming and shooting. As a result, the accessible control scheme simplified interaction but also changed the quality of player control. This trade-off is important when interpreting the evaluation results, as improvements in usability may occur alongside reductions in precision, agency, or perceived skill.

5.3 Gameplay Systems

5.3.1 Player

The player is implemented as a composition of independent systems, a full outline of which can be found in Appendix D, Table D.1. PlayerController acts as the connection between input handling and player systems. The controller implements the IInput interface and is responsible for retrieving input values and exposing them as general player actions. This means that player scripts do not need to know whether the player is using the standard or accessible control scheme; they only receive the resulting action values. As a result, control schemes can be swapped dynamically without affecting gameplay behaviour or responsiveness, supporting NFR-03. Both the standard and accessible schemes ultimately produce the same set of actions, ensuring that all gameplay systems operate consistently regardless of the active configuration. The action scripts (PlayerMove, PlayerAim, PlayerShoot, PlayerReload, PlayerDodge) process input through the PlayerController layer and apply corresponding forces or state changes to the player character.

This modular design allows individual mechanics to be isolated and tested independently. It also ensures that modifications to one aspect

of player behaviour do not introduce unintended effects elsewhere in the system. The players object structure can be seen in Appendix D, Figure D.1)

5.3.2 Enemies

As discussed in Section 4.2, enemy behaviour was intentionally simplified to maintain a controlled interaction environment and minimise confounding variables. However, during implementation and testing, this level of simplification introduced unintended limitations in gameplay dynamics. In practice, enemies exhibited highly predictable movement patterns and frequently clustered together, reducing the variability of encounters. While this supported experimental control, it also reduced gameplay challenge and limited opportunities for meaningful interaction. This was particularly significant given that player experience is a key component of the evaluation, as overly predictable behaviour diminished engagement and reduced the observable impact of different control schemes. To address this, additional behaviours were introduced in the form of separation and strafing. Separation ensured that enemies maintained distance from one another, preventing clustering and creating more distributed threats. Strafing introduced minor lateral movement, increasing variability in enemy positioning and requiring more responsive aiming and movement from the player.

Extending the baseline design, each enemy therefore consists of movement, collision, and health components (see Figure D.2), with movement directed towards the player and supplemented by separation and strafing behaviours. These behaviours represent a minimal but effective change increasing interaction variability without compromising the controlled conditions required for evaluation.

5.4 Metrics Collection System

To support the evaluation methodology, a dedicated metrics system was implemented to capture both performance and interaction data. An event-driven architecture was used to decouple data collection from gameplay logic. Gameplay events such as enemy elimination and collisions are broadcast through a central event system, allowing them to be recorded without modifying core gameplay scripts. Input data is captured separately through input tracking components, which record when inputs are initiated and released. These events are processed to derive higher-level interaction metrics such as sustained, repeated, and simultaneous inputs. This separation between gameplay and analytics ensures that data collection does not interfere with system behaviour. It also improves extensibility, allowing additional metrics to be introduced without modifying existing systems.

5.5 Trade-offs and Limitations

The implementation highlights several key trade-offs between accessibility, control, and gameplay experience. A primary challenge was maintaining a balance between reducing input complexity and preserving meaningful interaction. Automated mechanics reduce the need for precise and coordinated input, which can simplify interaction but also diminish player agency.

Additionally, the requirement for identical gameplay systems constrained the design of accessibility features. Rather than modifying mechanics, accessibility was implemented through input abstraction and automation. This limits the extent to which accessibility can be integrated into gameplay design. These solutions improved usability but introduced trade-offs in precision and control.

Beyond these design decisions, the choice of development platform also introduced its own constraints. The use of the Unity engine influenced how the system was designed and implemented. Unity runs on a continuous update loop, meaning certain behaviours, such as automated shooting and reloading, were implemented using repeated checks each frame. While this works well for the prototype, it may not fully reflect how such systems would be implemented in more complex triple-A games. These factors do not undermine the evaluation, but they do show that the prototype is influenced by how Unity works as a development tool. As a result, the findings should be understood within the context of this implementation.

Overall, the implementation successfully produced a controlled and functional prototype that supports systematic evaluation. The implementation process showed that accessibility features influence more than player input; they also affect how the system must be designed and implemented. Supporting different control schemes required input abstraction, separation between input and gameplay logic, and careful handling of automated actions. This reinforces the point raised in Section 2.3 that accessibility should be considered throughout the full development process rather than treated as an optional layer. Planning for accessibility from the start makes it easier to integrate alternative inputs consistently, avoid conflicts between mechanics, and preserve the underlying gameplay structure.

6 Results and Evaluation

6.1 Processing Data

Before analysis, raw gameplay metrics were exported from the prototype and organised by participant and control scheme. For each participant, data from the standard and accessible conditions were paired to support the within-subject approach discussed in Section 3.4.1. Objective metrics (Appendix H) were summarised using averages, percentage changes, and comparisons. Questionnaire responses were converted into numerical values using a five-point Likert scale, ranging from 1 for “Strongly Disagree” to 5 for “Strongly Agree”. Average responses were then calculated for each question across both control schemes. Time-based gameplay metrics were analysed by calculating average cumulative values across participants at regular points during each session. This allowed trends to be compared over time and helped identify when differences between the standard and accessible conditions emerged. Correlation analysis was also used to explore relationships between gameplay metrics. Reflecting on the limitations and aims explored in Section 3.4.3, these correlations are interpreted as exploratory indicators of possible relationships rather than statistically conclusive evidence. Overall, the analysis focuses on descriptive trends, comparative differences, and triangulation between objective metrics, questionnaire responses, and qualitative feedback. The full list of questionnaire and interview results can be found in Appendix I.

6.2 Participant Context

Eight participants completed the evaluation. Participants represented a range of gaming experience levels (Figure I.3), although most had some regular experience with digital games. Four participants reported playing games several times a week, three played a few times a month, and one played rarely. This supports the intended aim (discussed in Section 3.4.3) of capturing varied levels of familiarity with game controls, while also recognising that the sample cannot be considered representative of the wider gaming population or of players with motor impairments.

6.3 Reduction in Input Demand

Prior to assessing the research questions, it is important to examine the impact of the accessible control scheme on interaction demands to confirm its success. Figure 6.1 shows the total inputs for each participant across both conditions. Results show a substantial reduction for all participants, with an average decrease of approximately 85%.

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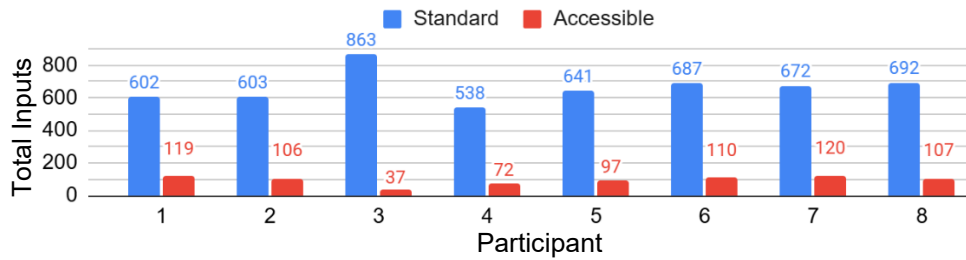


Figure 6.1: Total Inputs of Participants

Additionally, more complex forms of interaction were effectively eliminated. Appendix H, Table H.1 shows that simultaneous inputs, sustained inputs, and repeated inputs were almost entirely removed in the accessible version. This indicates that the control scheme reduced the quantity of interaction and simplified its structure.

Objective findings are supported by questionnaire results (Figure 6.2), which show consistent reductions in perceived input demand. Participants reported lower requirements for precision, accuracy, and timing in the accessible condition. The most substantial changes were observed in physical effort (4.00 to 1.88) and fatigue (4.13 to 2.13), indicating that the reduction in input complexity translated directly into a less demanding gameplay experience.

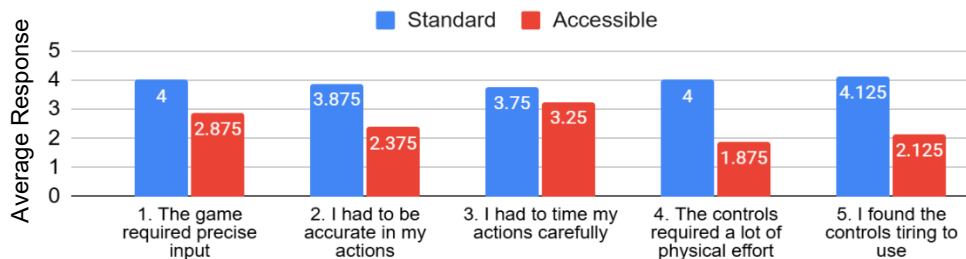


Figure 6.2: Input Demand Questionnaire Results

This was also reflected in observations during gameplay. Several participants visibly readjusted their hands or stretched their wrists during the standard condition, particularly after repeated shooting. Comments made during the think-aloud process also supported this finding, with participants describing the standard controls as “tiring”, and “difficult” to manage during more intense phases of play. These changes suggest a significant reduction in interaction burden. By minimising the need for complex coordinated inputs, the accessible control scheme lowered the physical and cognitive demands placed on the player. This directly supports the project's third objective to reduce input demand and complexity, ultimately decreasing fatigue and effort during gameplay.

6.4 Performance – RQ1

RQ1. What impact do software-based motor-accessibility features have on gameplay performance?

This section evaluates the impact of the accessible control scheme on gameplay performance. The results show that performance improved overall in the accessible condition. However, The improvement appears to reflect more than general learning or adaptation. Instead, the data suggests that reduced input demand changed how participants interacted with the game, enabling more frequent action and encouraging a more offensive playstyle.

6.4.1 Performance Metrics Results

The accessible control scheme improved overall gameplay performance. Every participant achieved a higher final score in the accessible condition than in the standard condition, as shown in Figure 6.3. A Wilcoxon signed-rank test [39] was used because the study compared paired results from the same participants across two conditions, and the small sample size made a non-parametric test more appropriate than assuming normality (see Figure H.3 for full results). Average score increased by 16.69%, indicating that this difference was statistically significant, $W = 0.0$, $p = 0.0078$. This provides strong evidence that the accessible scheme improved overall performance within this sample. However, the size of improvement varied between participants. While some participants showed relatively modest gains, others improved substantially, with the largest increase reaching approximately 50%. This suggests that the accessible scheme did not affect all participants equally, and that its impact may have depended on individual adaptation, prior gameplay experience, or playstyle.

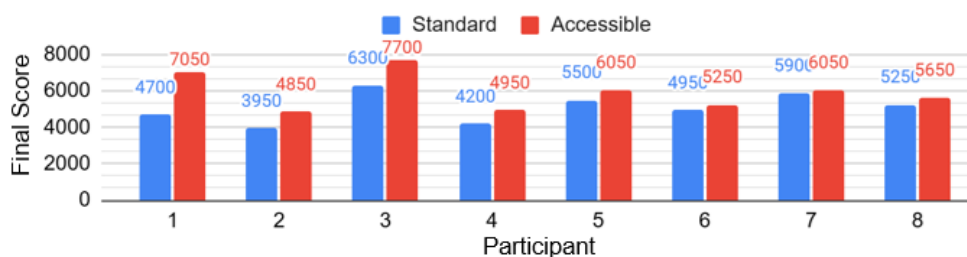


Figure 6.3: Final Score of Each Participant

The supporting performance metrics help explain the improvement in score. Enemy eliminations increased by 11.27% in the accessible condition, which was also statistically significant, $W = 0.0$, $p = 0.0178$. Deaths decreased by 34.62%, $W = 2.0$, $p = 0.0374$, suggesting that participants survived more effectively overall. Dodges performed also

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increased by 35.16%, $W = 0.0$, $p = 0.0176$, indicating that the simplified dodge input made defensive actions easier to execute. Enemy collisions decreased by 19.58%, but interestingly this reduction was not statistically significant, $W = 6.0$, $p = 0.1094$. This indicates that performance improvements were mainly linked to stronger offensive output and survivability, rather than consistent improvements in avoiding enemies.

Analysing average eliminations over time (Figure 6.4) provides additional context as to where these changes occur. During the early stages, both control schemes produced similar outcomes, suggesting that the standard controls were manageable when enemy pressure was low. As difficulty increased, differences between the conditions became more visible, with eliminations diverging after approximately 120 seconds. This aligns with the increase in enemy count, suggesting that the accessible scheme became more beneficial when there were more targets and greater pressure to perform repeated offensive actions. This supports the interpretation that performance gains were linked to sustained combat output rather than simply an immediate improvement in baseline ability.

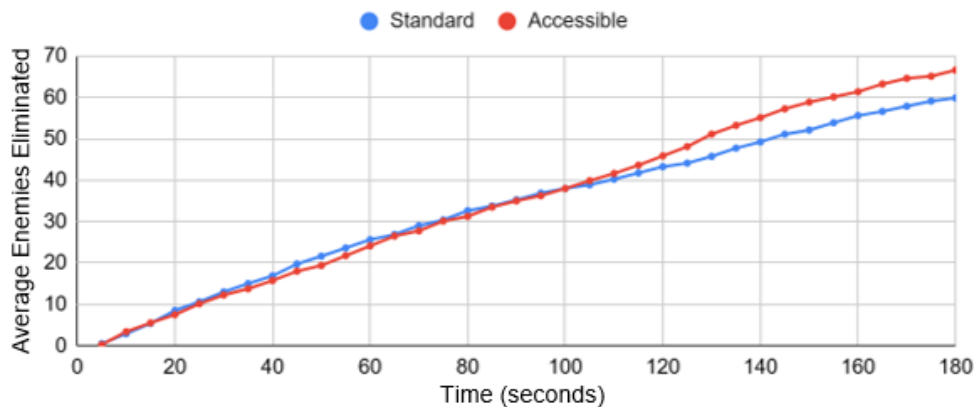


Figure 6.4: Average Enemies Eliminated Over Time

6.4.2 The Playstyle Shift

To explore why performance improved, Pearson correlation analysis was used to examine relationships between gameplay metrics. The main improvement appeared to be in offensive output. Scores and enemy eliminations increased significantly, while the reduction in collisions was not statistically significant. This suggests that the accessible version improved performance more reliably through attacking than through avoidance.

The correlation results (Tables 6.1 and 6.2) support this interpretation. In the standard condition, shots fired had little relationship with

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collisions, $r = 0.11$, whereas this increased to $r = 0.41$ in the accessible condition. This implies that higher offensive activity may have been associated with greater exposure to enemy contact. A similar pattern was seen between eliminations and collisions, which became positively correlated in the accessible condition. However, this did not necessarily lead to poorer survival. In the accessible condition, eliminations were negatively correlated with deaths, $r = -0.49$, suggesting that participants who eliminated more enemies tended to die less often. This appears counterintuitive, as greater enemy contact would normally be expected to increase the risk of death. However, it suggests that offensive output may have helped participants manage danger by reducing the number of active threats and sustaining health through elimination rewards.

Standard	Shots Fired	Accuracy	Collisions	Eliminations	Dodges	Deaths
Shots Fired	1.00	-0.70	0.11	0.91	-0.01	0.05
Accuracy	-0.70	1.00	-0.41	-0.50	0.10	-0.05
Collisions	0.11	-0.41	1.00	-0.05	0.03	0.72
Eliminations	0.91	-0.50	-0.05	1.00	-0.25	0.00
Dodges	-0.01	0.10	0.03	-0.25	1.00	-0.20
Deaths	0.05	-0.05	0.72	0.00	-0.20	1.00

Table 6.1: Correlations Between Standard Gameplay Metrics

Accessible	Shots Fired	Accuracy	Collisions	Eliminations	Dodges	Deaths
Shots Fired	1.00	-0.38	0.41	0.94	-0.16	-0.41
Accuracy	-0.38	1.00	-0.56	-0.08	0.14	-0.26
Collisions	0.41	-0.56	1.00	0.32	-0.84	0.32
Eliminations	0.94	-0.08	0.32	1.00	-0.19	-0.49
Dodges	-0.16	0.14	-0.84	-0.19	1.00	-0.37
Deaths	-0.41	-0.26	0.32	-0.49	-0.37	1.00

Table 6.2 Correlations Between Accessible Gameplay Metrics

Overall, these patterns suggest that input reduction alone does not fully explain the performance gains. The accessible scheme changed the relative cost and value of different actions, making attacking easier and encouraging a more offensive playstyle. This supports the argument made in Section 2.4 that accessibility features can influence gameplay strategy, rather than only reducing input complexity. However, this shift cannot be fully understood without considering the wider design of the prototype.

6.4.3 Interaction with Game Design

Accessibility features did not operate in independently; their effects depended on the game system in which they were implemented. In this prototype, automated shooting reduced the effort required to perform offensive actions. At the same time, the scoring system

strongly rewarded eliminations, with each enemy eliminated awarding 100 points and contributing to health recovery, while collisions carried a smaller 50-point penalty. This meant that the accessible scheme made the most rewarded action easier to perform, while the cost of enemy contact remained relatively manageable.

This helps explain why performance improved more clearly through eliminations than through avoidance. The accessible scheme amplified an offensive incentive already present in the game design. Participants could achieve higher scores by eliminating enemies more consistently, even where avoidance did not improve to the same extent. This may have reduced the relative importance of precise movement and avoidance within the accessible condition. These findings highlight the importance of considering accessibility features alongside mechanics, scoring, and balance. The accessible control scheme reduced motor demand, but its effect on performance was shaped by the surrounding game system. This reinforces the wider argument explored in Section 2.3 that accessibility should be considered as part of core game design, rather than as an isolated feature added after the main mechanics have been defined.

6.4.4 Performance Conclusion

In response to RQ1, software-based motor-accessibility features improved performance in this prototype, but they also changed how participants interacted with the game. By reducing the effort required to attack, the accessible scheme encouraged a more offensive playstyle, which was further reinforced by the prototype's scoring and reward systems. Therefore, performance improvements can be understood because of how accessibility features interacted with the wider game design.

6.5 Usability – RQ2

RQ2. How do software-based motor-accessibility features affect usability?

Section 6.4 established that the accessible control scheme significantly reduced input demand across total, simultaneous, sustained, and repeated inputs. RQ2 builds on this by examining whether these reductions translated into improved usability. Usability is assessed through questionnaire responses, qualitative feedback, and observations made during playtesting.

6.5.1 Perceived Usability and Learnability

Figure 6.5 shows usability questionnaire responses. Participants reported a clear improvement in ease of use, (3.00 to 4.00). Perceived complexity also decreased substantially (4.13 to 1.63). This indicates

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that participants experienced the accessible control scheme as much simpler to operate. These findings align with the input demand results presented earlier, suggesting that the measurable reduction in required inputs was also reflected in participants' subjective experience.

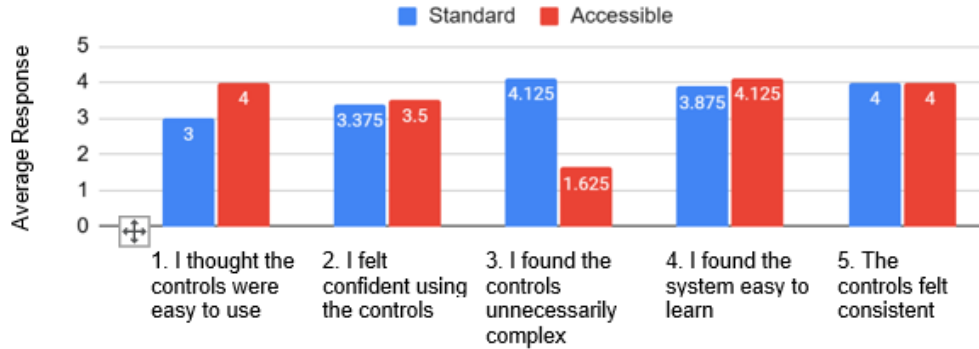


Figure 6.5: Average Responses of Usability Questions

Learnability showed a slight improvement in the accessible condition, which could imply that the automated features and simplified layout allowed participants to adapt with limited exposure. However, considering the limitations discussed in section 3.4.1, learning bias may have influenced results. To mitigate this, interview feedback supports these findings. Participants frequently described the accessible version as “simpler”, “easier to use”, and less demanding. Confidence and consistency remained relatively stable across both conditions, indicating that the simplified controls did not prevent participants from performing the main gameplay actions. This suggests that usability improved without making the accessible scheme confusing or unreliable. Several participants also stated that they could focus more on the game itself rather than on managing the controls. This signifies how usability was improved by reducing the attention required to operate the control scheme.

6.5.2 Coordination and Interaction

The main usability benefit of the accessible scheme was the reduction in coordination required during play. In the standard condition, participants had to manage movement, aiming, shooting, dodging, and reloading across both hands. Feedback indicated that this became difficult during more intense phases, particularly when repeated shooting had to be combined with movement and avoidance. In contrast, the accessible scheme simplified the interaction structure by reducing the number of actions that had to be manually controlled at the same time. Automated shooting and reloading removed repetitive actions, while the one-handed design was praised by participants for reducing the need to coordinate keyboard and mouse inputs. This

made the control scheme easier to manage during gameplay, allowing participants to focus more on enemies, positioning, and survival. Ultimately, the accessible scheme improved usability by reducing interaction management. The benefit was that participants had fewer competing actions to coordinate under pressure.

6.5.3 Usability Trade-off

Despite these usability improvements, the accessible control scheme introduced trade-offs. Some participants reported that the standard controls felt more natural because they were familiar with conventional keyboard-and-mouse layouts. In contrast, the accessible version was easier to operate but sometimes felt less precise. Interestingly, the use of shared mouse input for both movement and targeting also introduced issues. While this reduced coordination as previously discussed, it also limited the ability to move in one direction while aiming or attacking in another. Participants noted that this made certain movements feel less controlled, particularly when trying to reposition quickly while continuing to attack. This shows that reducing input complexity can improve usability while also reducing control flexibility.

These trade-offs show that usability depends on more than how easy a control scheme is to operate. A usable accessibility feature must also support the player's intentions clearly. In this case, the accessible scheme reduced effort and complexity but also introduced limitations in precision and movement control.

6.5.4 Usability Conclusion

In response to RQ2, the accessible control scheme improved usability by reducing perceived complexity, improving ease of use, and making interaction easier to manage during gameplay. These improvements were closely linked to the reduction in input demand established in Section 6.4. However, the accessible scheme also introduced trade-offs in smoothness, precision, and control flexibility. This suggests that motor-accessibility features should aim to preserve clear control over player actions.

6.6 Player Experience

RQ3. What impact do software-based motor-accessibility features have on player experience?

While RQ2 focused on usability, RQ3 considers how the accessible control scheme affected the wider gameplay experience, including enjoyment, frustration, challenge, skill, and concentration. The results suggest that the accessible control scheme changed the gameplay experience rather than simply making the same experience easier to

access. Instead, it changed the nature of the experience by shifting the balance between comfort, challenge, control, and agency.

6.6.1 Comfort and Frustration

Figure 6.6 shows the average responses for experience questions. The accessible control scheme improved several aspects of player comfort. Average enjoyment increased slightly in the accessible condition, while frustration decreased. This suggests that reducing the physical and interaction demands of the control scheme made the game more manageable and less frustrating for participants.

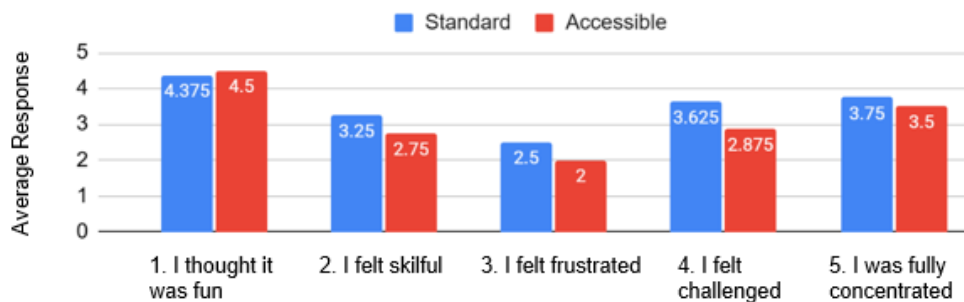


Figure 6.6: Average Response to Experience Questions

Qualitative feedback supported this interpretation. As previously mentioned, several participants stated that the accessible version allowed them to focus more on the game itself rather than on managing the controls. For example, participants described being able to focus more on what was happening on screen, killing enemies, and surviving, rather than dealing with repeated clicking or coordinating multiple inputs. Ultimately, the accessible control scheme reduced the amount of attention required for basic interaction, allowing participants to engage more directly with the gameplay situation.

However, this also changed the tone of the experience. Some participants described the accessible version as less intense or less pressured than the standard condition. For some, this was positive, as it made the game feel more relaxed and manageable. Therefore, the accessible scheme improved comfort by reducing physical strain and the stress associated with keeping up with complex controls.

6.6.2 Challenge, Skill, and Engagement

Although a more comfortable and relaxed tone was received positively by some, the accessible control scheme also reduced some aspects of challenge and skill expression for others. Overall, perceived challenge, skill, and concentration were reported lower in the accessible condition. This suggests that while the accessible scheme made gameplay easier to manage, it also reduced the level of effort and focus required to perform successfully.

This was reflected in participant feedback. Some participants felt that the accessible version was less demanding because there was less to do, while others described it as less intense or less involving. These comments imply that the accessible scheme reduced unnecessary difficulty while also changing how challenge was experienced. By automating shooting and simplifying other actions, the game required less precise coordination and fewer active decisions during gameplay. These findings point to an important distinction. Improved accessibility and usability do not automatically result in a more engaging experience for every player. In this study, the accessible control scheme helped participants perform more comfortably, but it also reduced some of the challenge and skill expression associated with the standard controls. As a result, the accessible version shifted the experience from a more intense and physically demanding action game towards a more relaxed and automated form of play.

This supports the argument made in Section 2.4 that accessibility design can influence the wider experience of play. Even when the underlying game mechanics remain the same, altering how players perform actions can change how challenge, control, and engagement are experienced. For developers, this suggests that accessibility features should be considered in terms of how they affect the intended feel of the game. This raises an important design question: how can input demands be reduced while preserving a comparable sense of agency, challenge, and accomplishment?

6.6.3 Player Experience Conclusion

In response to RQ3, the accessible control scheme improved comfort and reduced frustration but also changed the type of gameplay experience produced. For several participants, the accessible version was more manageable, less stressful, and easier to engage with. However, these benefits were accompanied by reductions in perceived challenge, skill expression, intensity, and direct control. This shows that software-based motor-accessibility features can affect usability, performance, and the wider experience of play. They can reshape it by changing how challenge, agency, and engagement are distributed. The accessible scheme therefore created a trade-off: it reduced barriers and improved comfort but also made the game feel less intense and less directly controlled for some participants. This reinforces the need to design accessibility features as part of the overall gameplay experience, rather than treating them only as input reductions.

7 Conclusion

This project investigated how software-based motor-accessibility features affect gameplay performance, usability, and player experience within a controlled game prototype. To achieve this, research and industry guidance on motor accessibility were synthesised, a 2D action-game prototype was designed and implemented, and standard and accessible control schemes were evaluated using gameplay metrics, questionnaire responses, observation, and qualitative feedback. The project met its objectives by translating accessibility guidance into a working system and using it to examine the practical effects of accessibility-informed input design. The prototype successfully elicited common motor demands and included an accessible control scheme driven by research to mitigate these demands. Through simplified input, automation, and assistance, the same gameplay functions were compared under different interaction conditions.

The findings suggest that software-based motor-accessibility features can reduce input demand, improve usability, and support improved gameplay performance. However, the results also show that these features affect more than control difficulty. By changing how actions are performed, the accessible scheme altered how participants interacted with the game, encouraging a more offence-focused playstyle and changing the balance between comfort, challenge, skill, and control. This represents the main contribution of the project. It demonstrates that accessibility features should be considered as design choices that interact with mechanics, scoring, balance, and player experience. The project therefore contributes exploratory evidence that motor-accessibility features can improve interaction, while raising important design questions: how can input demand be reduced without reducing agency, challenge, or the intended feel of play? How can accessibility features support different players' needs while preserving meaningful interaction?

As participants were not recruited based on motor impairment, the findings should be interpreted as exploratory evidence of design impact rather than conclusive validation of accessibility effectiveness for motor-impaired players. Future work should therefore evaluate similar systems with motor-impaired participants, larger samples, and more varied game genres. Overall, this project shows that software-based motor-accessibility features can reduce barriers to interaction while also affecting how the game is played and experienced. Accessibility must be considered as part of the wider game design process, because changing how players interact with a game can also change how the game is played and experienced.

Appendix A: Requirements

ID	Type	Requirement	Severity
FR-01	Functional	The game must provide a 2D top-down environment that can be navigated	Must
FR-02	Functional	The game must include a controllable player character	Must
FR-03	Functional	The player must be able to move within the environment	Must
FR-04	Functional	The player must be able to aim	Must
FR-05	Functional	The player must be able to shoot	Must
FR-06	Functional	The player must be able to reload	Must
FR-07	Functional	The player must be able to dodge	Must
FR-08	Functional	The game must include enemy entities	Must
FR-09	Functional	Enemies must move towards the player	Must
FR-10	Functional	Enemies must take damage from shots, and be eliminated after too much damage is inflicted	Must
FR-11	Functional	Enemies must inflict damage when they collide with the player	Must
FR-12	Functional	The game must include a score tracker that increases with enemy eliminations and decreases with enemy collisions	Must
NFR-01	Non-Functional	The game environment should minimise visual, auditory, and environmental complexity	Should

Table A.1: Gameplay Requirements

ID	Type	Requirement	Severity
FR-13	Functional	The game must provide an alternative control scheme	Must
FR-14	Functional	The game must provide a feature to reduce sustained inputs	Must
FR-15	Functional	The game must provide a feature to reduce precision inputs	Must
FR-16	Functional	The game must provide a feature to reduce simultaneous inputs	Must
FR-17	Functional	The game must provide a feature to reduce rapid repeated inputs	Must
NFR-02	Non-Functional	Accessibility features must not alter core gameplay mechanics or objectives	Must
NFR-03	Non-Functional	Accessibility features should reduce motor effort without negatively impacting playability	Should
NFR-04	Non-Functional	The game should maintain consistent responsiveness across both control schemes	Should

Table A.2: Accessibility Requirements

Appendix B: Questionnaire and Interview Items

(CORGIS) Input Demand - RQ1, RQ2

1. The game required precise input
2. I had to be accurate in my actions
3. I had to time my actions carefully
4. The controls required a lot of physical effort
5. I found the controls tiring to use

(SUS) Usability - RQ2

1. I thought the controls were easy to use
2. I felt confident using the controls
3. I found the controls unnecessarily complex
4. I found the system easy to learn
5. The controls felt consistent

(GEQ) Experience - RQ3

1. I thought it was fun
2. I felt skilful
3. I felt frustrated
4. I felt challenged
5. I was fully concentrated

Figure B.1 Post-Session Questionnaire

Overall Experience - RQ3

How did you find the overall gameplay experience?

Control Differences - RQ2

What differences did you notice between the two control schemes?

Motor Effort - RQ2

Did one version feel more physically demanding than the other? Why?

Usability - RQ2

Were there any controls you found difficult or frustrating? Why?

Accessibility – RQ2

Do you think the accessible version would benefit players with limited motor ability? Why?

Trade-Offs - RQ3

Did the accessible version change the gameplay experience in any way (positively or negatively)?

Figure B.2: Post-Playtest Interview Questions

Appendix C: Requirements-to-Design Mapping

ID	Requirement	Design
FR-01	The game must provide a 2D top-down environment that can be navigated	Use a 2D top-down camera perspective and provide a grid to navigate.
FR-02	The game must include a controllable player character	Create an input system that can be mapped to perform player actions
FR-03	The player must be able to move within the environment	Apply directional movement force based on a vector input
FR-04	The player must be able to aim	Apply rotational force based on mouse input
FR-05	The player must be able to shoot	Instantiate and shoot projectiles from the player based on combat input
FR-06	The player must be able to reload	Create an ammo system, decreasing with each shot, and replenishing with reload input
FR-07	The player must be able to dodge	Apply directional dodge force to the player based on dodge input
FR-08	The game must include enemy entities	Create an enemy spawner, that instantiates enemies over time.
FR-09	Enemies must move towards the player	Apply directional movement force based on the position of the player
FR-10	Enemies must take damage from shots, and be eliminated after too much damage is inflicted	Add health to enemies, decreasing from shot collisions, being eliminated after health is depleted.
FR-11	Enemies must inflict damage when they collide with the player	Add health to the player, decreasing on enemy collisions, dying after health is depleted
FR-12	The game must include a score tracker that increases with enemy eliminations and decreases with enemy collisions	Create a tracker that increases with enemy eliminations, and decreases with enemy collisions

Table C.1: Design of Gameplay Requirements

ID	Requirement	Design
FR-13	The game must provide an alternative control scheme	Combine accessible features into a scheme that can be used instead of a standard scheme
FR-14	The game must provide a feature to reduce sustained inputs	Using click to move instead sustained WASD movement inputs
FR-15	The game must provide a feature to reduce precision inputs	Automate shooting to remove the need to multitask clicking and moving the mouse at the same time
FR-16	The game must provide a feature to reduce simultaneous inputs	Using click to move instead simultaneous WASD inputs, as well as all inputs only requiring the mouse instead of both mouse and keyboard inputs simultaneously
FR-17	The game must provide a feature to reduce rapid repeated inputs	Map the double-tap dodge to a single input, and automate shooting and reloading to remove repetition

Table C.2: Design of Accessibility Requirements

Appendix D: Game Implementation

Requirement	Implementation	Description
FR-02 (Controllable Character)	PlayerController.cs	Inherits the IInput interface and makes use of an action map, connecting input to actions.
FR-03 (Move)	PlayerMove.cs	Takes directional input as a vector and applies a corresponding movement force
FR-04 (Aim)	PlayerAim.cs	Takes directional input as a quaternion and applies a rotation
FR-05 (Shoot)	PlayerShoot.cs	Takes a Boolean input, tells the Weapon to shoot when true.
FR-06 (Reload)	PlayerReload.cs	Takes a Boolean input, tells the Weapon to reload when true.
FR-05 and FR-06 (Combat)	Weapon.cs	Shoots projectiles and handles reloading based on PlayerShoot.cs and PlayerReload.cs.
FR-07 (Dodge)	PlayerDodge.cs	Takes a Boolean input, applying an impulse force when true.

Table D.1: Player Scripts

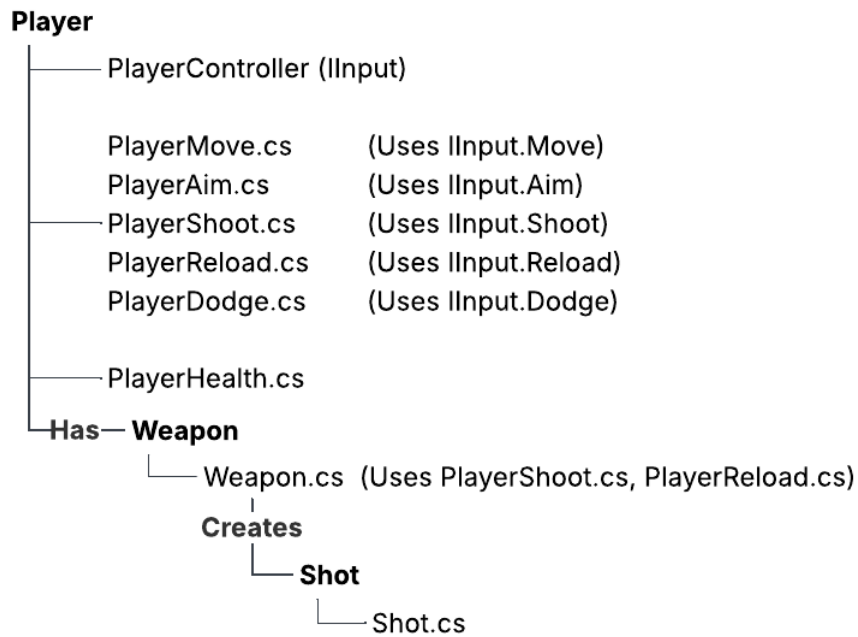


Figure D.1: Player Structure

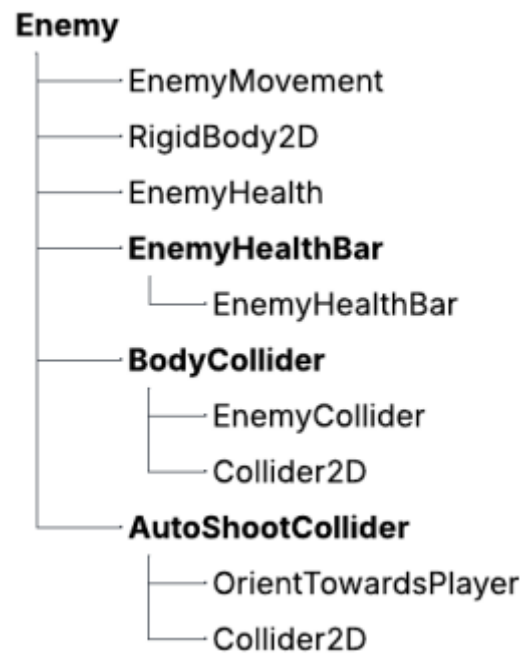


Figure D.2: Enemy Structure

Appendix E: Input System Action Maps

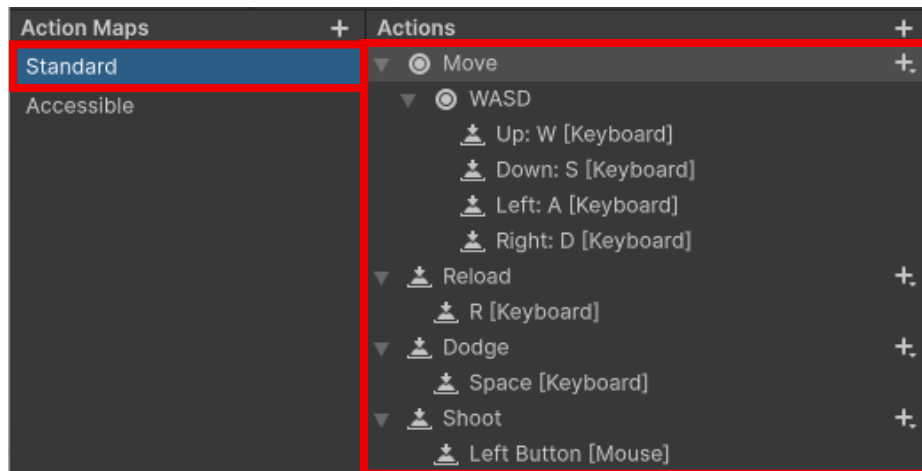


Figure E.1: Standard Control Scheme Action Map

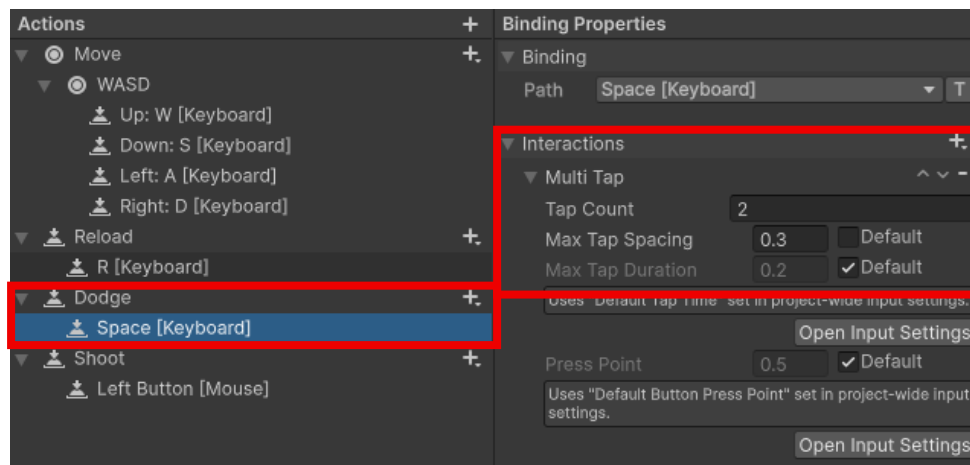


Figure E.2: Standard Control Scheme Multi-Tap Dodge Input

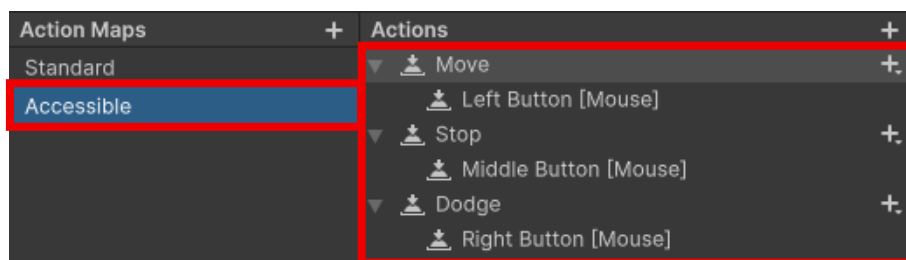


Figure E.3: Accessible Control Scheme Action Map

Appendix F: Key Code Snippets

```
public interface IInput
{
    public Vector2 Move(); // Corresponds to FR-03
    public Quaternion Aim(); // Corresponds to FR-04
    public bool Shoot(); // Corresponds to FR-05
    public bool Reload(); // Corresponds to FR-06
    public bool Dodge(); // Corresponds to FR-07
}
```

Figure F.1: IInput Interface

```
private bool IsAimingAtEnemy()
{
    Collider2D hit = Physics2D.OverlapPoint(
        MousePosition,
        _enemyBoxLayer);

    return hit != null;
}
```

Figure F.2: AccessiblePlayerInput.cs IsAimingAtEnemy() Function

```
private bool IsAmmoEmpty()
{
    return _weapon != null && _weapon.AmmoCount <= 0;
}
```

Figure F.3: AccessiblePlayerInput.cs IsAmmoEmpty() Function

Appendix G: Gameplay Screenshots and Recordings

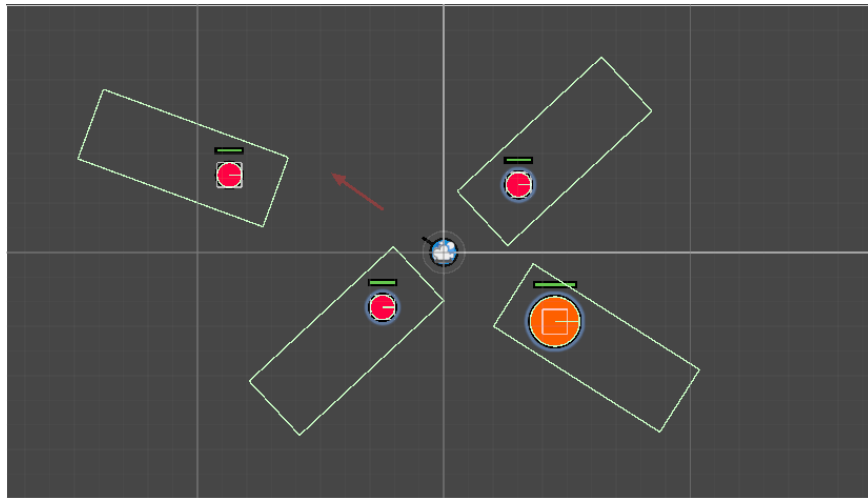


Figure G.1: AutoShoot Hitboxes

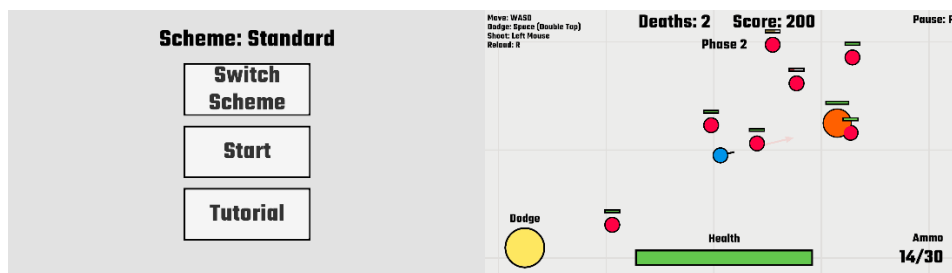


Figure G.2: Standard Scheme

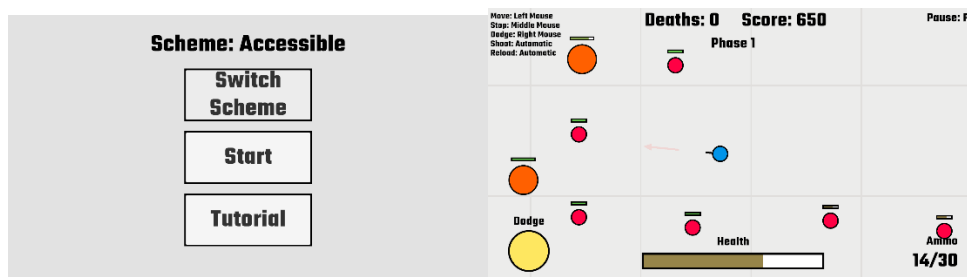


Figure G.3: Accessible Scheme

https://drive.google.com/drive/folders/1TlzTlwGnUs2AWfrNSr-QRfUecu4FAWnd?usp=drive_link

Figure G.4: Playtest Screen Recordings

Appendix H: Gameplay and Input Metrics

Participant	Total Inputs		Simultaneous Inputs		Sustained Inputs		Repeated Inputs	
	Standard	Accessible	Standard	Accessible	Standard	Accessible	Standard	Accessible
1	602	119	260	0	95	0	166	3
2	603	106	222	0	79	0	235	1
3	863	37	339	0	72	0	624	0
4	538	72	158	2	76	0	215	0
5	641	97	211	0	83	0	252	1
6	687	110	259	2	67	0	277	5
7	672	120	273	0	71	0	272	3
8	692	107	299	1	83	0	291	1

Table H.1: Input Metrics Results

	A	B	C	D	E	F	G	H	I
1		Score		Collisions		Eliminations		Deaths	
2	Participant	Standard	Accessible	Standard	Accessible	Standard	Accessible	Standard	Accessible
3	1	4700	7050	10	15	52	78	1	1
4	2	3950	4850	21	9	50	53	3	2
5	3	6300	7700	18	16	72	85	3	1
6	4	4200	4950	22	19	53	59	5	3
7	5	5500	6050	18	15	64	68	3	4
8	6	4950	5250	19	15	59	60	5	3
9	7	5900	6050	14	11	66	66	3	1
10	8	5250	5650	21	15	63	64	3	2
11	Average	5093.75	5943.75	17.875	14.375	59.875	66.625	3.25	2.125
12									
13		Shots Fired		Accuracy		Dodges Performed			
14	Participant	Standard	Accessible	Standard	Accessible	Standard	Accessible		
15	1	286	452	87.76	81.86	13	14		
16	2	304	317	74.67	76.34	14	18		
17	3	517	578	65.96	67.82	12	16		
18	4	305	385	78.36	68.31	8	14		
19	5	372	410	80.65	76.34	9	15		
20	6	359	379	86.63	74.67	14	14		
21	7	370	396	78.65	81.31	8	17		
22	8	382	369	78.01	78.59	13	15		
23	Average	361.875	410.75	78.84	75.66	11.375	15.375		

Table H.2: Gameplay Metrics

Metric	Standard mean	Accessible mean	% change	Wilcoxon W	p-value
Score	5093.75	5943.75	16.69%	0	0.0078
Deaths	3.25	2.12	-34.62%	2	0.0374
Enemy collisions	17.88	14.38	-19.58%	6	0.1094
Enemies eliminated	59.88	66.62	11.27%	0	0.0178
Dodges performed	11.38	15.38	35.16%	0	0.0176
Total inputs	662.25	96	-85.50%	0	0.0078
Simultaneous inputs	252.62	0.62	-99.75%	0	0.0078
Sustained inputs	78.25	0	-100.00%	0	0.0078
Repeated inputs	291.5	1.75	-99.40%	0	0.0078

Table H.3 Wilcoxon Signed-rank Test Results

Appendix I: Questionnaire and Interview Results

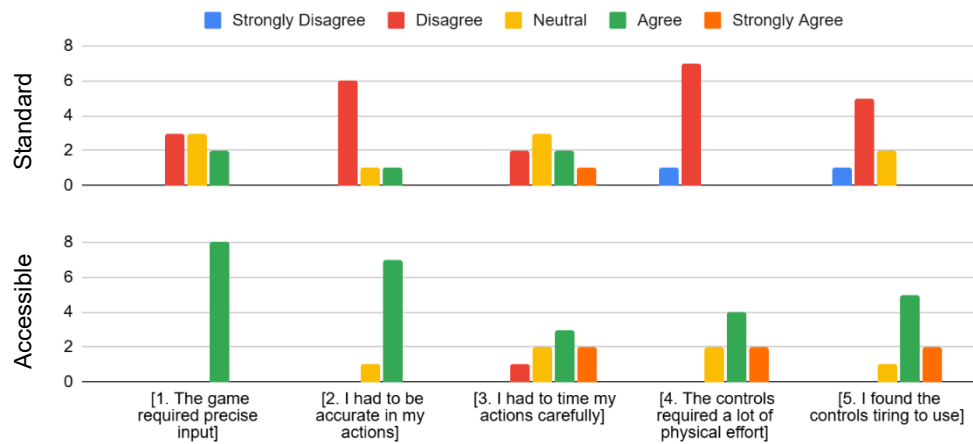


Figure I.1: Input Demand Questionnaire Results

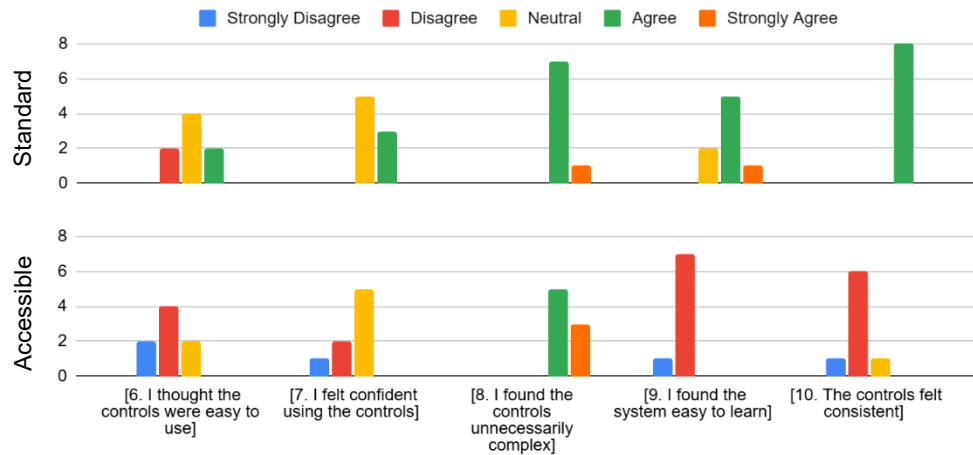


Figure I.2: Usability Questionnaire Results

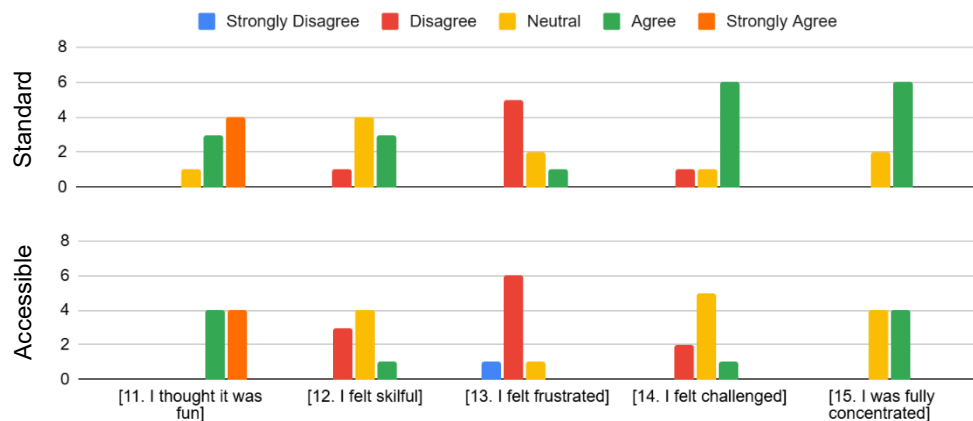


Figure I.3: Experience Questionnaire Results

Appendix I: Questionnaire and Interview Results

Participant	[1. The game required precise input]	[2. I had to be accurate in my actions]	[3. I had to time my actions carefully]	[4. The controls required a lot of physical effort]	[5. I found the controls tiring to use]
1	Agree	Agree	Neutral	Neutral	Agree
2	Agree	Neutral	Disagree	Neutral	Neutral
3	Agree	Agree	Strongly Agree	Strongly Agree	Strongly Agree
4	Agree	Agree	Strongly Agree	Agree	Strongly Agree
5	Agree	Agree	Agree	Agree	Agree
6	Agree	Agree	Agree	Agree	Agree
7	Agree	Agree	Agree	Strongly Agree	Agree
8	Agree	Agree	Neutral	Agree	Agree
Participant	[6. I thought the controls were easy to use]	[7. I felt confident using the controls]	[8. I found the controls unnecessarily complex]	[9. I found the system easy to learn]	[10. The controls felt consistent]
1	Neutral	Agree	Agree	Agree	Agree
2	Agree	Agree	Agree	Strongly Agree	Agree
3	Disagree	Neutral	Agree	Agree	Agree
4	Disagree	Neutral	Agree	Neutral	Agree
5	Neutral	Neutral	Agree	Agree	Agree
6	Agree	Agree	Strongly Agree	Agree	Agree
7	Neutral	Neutral	Agree	Agree	Agree
8	Neutral	Neutral	Agree	Neutral	Agree
Participant	[11. I thought it was fun]	[12. I felt skillful]	[13. I felt frustrated]	[14. I felt challenged]	[15. I was fully concentrated]
1	Strongly Agree	Agree	Disagree	Disagree	Agree
2	Agree	Neutral	Disagree	Neutral	Agree
3	Strongly Agree	Agree	Agree	Agree	Agree
4	Agree	Disagree	Neutral	Agree	Neutral
5	Strongly Agree	Neutral	Disagree	Agree	Agree
6	Strongly Agree	Agree	Disagree	Agree	Agree
7	Neutral	Neutral	Disagree	Agree	Neutral
8	Agree	Neutral	Neutral	Agree	Agree

Table I.1: Standard Version Participant Questionnaire Results

Participant	[1. The game required precise input]	[2. I had to be accurate in my actions]	[3. I had to time my actions carefully]	[4. The controls required a lot of physical effort]	[5. I found the controls tiring to use]
1	Agree	Neutral	Disagree	Strongly Disagree	Disagree
2	Neutral	Disagree	Neutral	Disagree	Disagree
3	Agree	Agree	Strongly Agree	Disagree	Disagree
4	Neutral	Disagree	Disagree	Disagree	Neutral
5	Disagree	Disagree	Agree	Disagree	Strongly Disagree
6	Disagree	Disagree	Neutral	Disagree	Neutral
7	Neutral	Disagree	Agree	Disagree	Disagree
8	Disagree	Disagree	Neutral	Disagree	Disagree
Participant	[6. I thought the controls were easy to use]	[7. I felt confident using the controls]	[8. I found the controls unnecessarily complex]	[9. I found the system easy to learn]	[10. The controls felt consistent]
1	Strongly Agree	Neutral	Disagree	Agree	Agree
2	Neutral	Neutral	Disagree	Agree	Agree
3	Strongly Agree	Strongly Agree	Strongly Disagree	Strongly Agree	Strongly Agree
4	Agree	Agree	Strongly Disagree	Agree	Agree
5	Neutral	Neutral	Disagree	Agree	Agree
6	Agree	Neutral	Disagree	Agree	Agree
7	Agree	Agree	Strongly Disagree	Agree	Agree
8	Agree	Neutral	Disagree	Agree	Neutral
Participant	[11. I thought it was fun]	[12. I felt skillful]	[13. I felt frustrated]	[14. I felt challenged]	[15. I was fully concentrated]
1	Strongly Agree	Disagree	Disagree	Disagree	Agree
2	Agree	Disagree	Strongly Disagree	Disagree	Agree
3	Strongly Agree	Agree	Disagree	Neutral	Agree
4	Strongly Agree	Neutral	Neutral	Agree	Neutral
5	Agree	Disagree	Disagree	Neutral	Agree
6	Agree	Neutral	Disagree	Neutral	Neutral
7	Strongly Agree	Neutral	Disagree	Neutral	Neutral
8	Agree	Neutral	Disagree	Neutral	Neutral

Table I.2: Accessible Version Participant Questionnaire Results

Appendix I: Questionnaire and Interview Results

	Standard Version									Accessible Version										
Participant	1	2	3	4	5	6	7	8		1	2	3	4	5	6	7	8			
Input Demand	Average									Average										
1. The game required precise input	4	4	4	4	4	4	4	4	4	4	3	4	3	2	2	3	2			
2. I had to be accurate in my actions	4	3	4	4	4	4	4	4	3.875	3	2	4	2	2	2	2	2			
3. I had to time my actions carefully	3	2	5	5	4	4	4	3	3.75	2	3	5	2	4	3	4	3			
4. The controls required a lot of physical effort	3	3	5	4	4	4	5	4	4	1	2	2	2	2	2	2	2			
5. I found the controls tiring to use	4	3	5	5	4	4	4	4	4.125	2	2	2	3	1	3	2	2			
Usability	Average									Average										
1. I thought the controls were easy to use	3	4	2	2	3	4	3	3	3	5	3	5	4	3	4	4	4			
2. I felt confident using the controls	4	4	3	3	3	4	3	3	3.375	3	3	5	4	3	3	4	3			
3. I found the controls unnecessarily complex	4	4	4	4	4	5	4	4	4.125	2	2	1	1	2	2	1	2			
4. I found the system easy to learn	4	5	4	3	4	4	4	3	3.875	4	4	5	4	4	4	4	4			
5. The controls felt consistent	4	4	4	4	4	4	4	4	4	4	4	5	4	4	4	4	3			
Experience	Average									Average										
1. I thought it was fun	5	4	5	4	5	5	3	4	4.375	5	4	5	5	4	4	5	4			
2. I felt skillful	4	3	4	2	3	4	3	3	3.25	2	2	4	3	2	3	3	3			
3. I felt frustrated	2	2	4	3	2	2	2	3	2.5	2	1	2	3	2	2	2	2			
4. I felt challenged	2	3	4	4	4	4	4	4	3.625	2	2	3	4	3	3	3	2.875			
5. I was fully concentrated	4	4	4	3	4	4	3	4	3.75	4	4	4	3	4	3	3	3			

Table I.3: Synthesised Questionnaire Results

How did you find the overall gameplay experience?

1. Overall I enjoyed the game it was quite intense but in a fun way.
2. It was an exciting experience, I liked how it got more hectic as it went on.
3. the game was fun and i enjoyed how it got harder as the phases progress - it felt challenging yet still rewarding
4. I liked it. It was pretty fun and chaotic.
5. I thought it was pretty fun. I liked the different phases and how it got harder.
6. it was good I found it entertaining and fun overall
7. It was fun I enjoyed shooting and dodging around enemies
8. I enjoyed it, It got more fun as I got to later phases

What differences did you notice between the two control schemes?

1. Found the accessible scheme layout far simpler, I enjoyed only needing one hand to play.
2. The standard one felt more natural to me but the accessible one was a lot easier to use.
3. i had more time to focus on the screen and what was actually happening in the game with version 2 - i felt like i had more control because i was less tired trying to keep up with clicking and i could just focus on playing the game
4. The first one was quite difficult I couldn't really do a lot of what I wanted to do at the same time. Whereas the second one made it easier to do more things at once.
5. I found myself having to put in more effort with the first one because the controls were more complex, the second one was very simple.
6. the second one was easier to use than the first and I didn't need to press as many things which was nice
7. The first one was more complicated, although it used the normal controls it was a bit overwhelming at times, the second one was simpler I could concentrate on the game more instead of the controls.
8. The controls of the second one were a lot simpler and there was more automatic controls in place instead of lots of different buttons

Did one version feel more physically demanding than the other? Why?

1. The standard version felt more demanding because of the coordination of both switching between keys on one hand, and aiming and shooting with the other.
2. Most of the difference came from the moving around and shooting, having to click for every bullet in the standard one was a bit repetitive, and the WASD movement was more demanding on my fingers because of the switching directions but it felt smoother and more like I was in control.
3. version 1 felt much more physically demanding than version 2 because i had to coordinate using 2 hands and sometimes felt jumbled or forgot that i could do certain things (like dodge) and my wrist started getting tired with the constant clicking to shoot
4. The first one felt a lot more physically demanding. I didn't like the shooting I struggled with aiming and shooting at the same time while also trying to move around I had to stop to shoot and aim alot.
5. The first one because I had to do lots of different things at the same time, whereas with the second one I only really had to press a single button every so often.
6. the first one felt more demanding because I had to multitask doing multiple things at the same time. With the second one I didn't have to click and hold so many things to move and shoot
7. The first one was more demanding because I had to juggle a lot of different buttons at the same time, and it was really repetitive as I played more.
8. The first one because my right wrist ached after a while from the constant clicking while also trying to aim

Were there any controls you found difficult or frustrating in any of the versions? Why?

1. In the standard version the dodge being a double tap was annoying and felt unnecessary, it felt like I had to go out of the way to ensure I did it right which took my focus away from doing other things. In the accessible

Appendix I: Questionnaire and Interview Results

- version I had trouble having to balance my aiming for moving and shooting, sometimes if I wanted to move in a certain direction quickly but shoot in another direction it was a bit difficult.
- Nothing was particularly difficult or frustrating, the only small thing was the dodge in the standard version was slightly harder to do because it was hard to tap my thumb quickly while also moving my other fingers.
 - the constant clicking to shoot became tiresome and started to hurt my wrist a little - it also made it a bit harder to focus on the other controls, my other hand and the games itself as it started to speed up and get faster
 - The shooting in the first one was a lot more difficult because I found it hard trying to shoot and move at the same time.
 - The shooting in the first one was pretty difficult because it became tiring towards the end I didn't want to keep clicking anymore. Also I didn't use the dodge in the first one until later because I didn't want to have to double click the space while moving it seemed a bit out of the way.
 - in the first one shooting was difficult, I liked the second one because I didn't even have to click anything. In the second one the movement did feel a bit weird because I felt like I didn't really have as much control over what I was doing.
 - The only thing I found a bit annoying was the shooting and dodging in the first version because it felt quite repetitive and tiring having to keep pressing it over and over again.
 - The shooting in the first one as well as the dodging because I had to kind of had to just keep spamming it the whole time and it was difficult to control while also trying to move

Do you think the accessible version would benefit players with limited motor ability?

- I think it would benefit someone that can't coordinate as many things at once. I was able to do relatively well with it and pick it up ok considering I wasn't fully adapted to it, so I think with practice it would work well because it is noticeable simpler.
- I think so, I can see how the standard version would be quite a lot so the accessible one could be quite good for that.
- yes - it was a lot less physically demanding. i already had to concentrate a bit more than normal playing version 1 even without mobility issues so i imagine that it would feel even more frustrating trying to keep up with some of the controls (like clicking a lot to shoot or having to use buttons like wasd and space that are somewhat far apart from each other) if i had mobility impairments
- Yes I do. Because it allowed me to do more of what I wanted to do without needing to press so many different buttons to do it, so I think someone that can't press as many things would benefit a lot.
- I think it would benefit them because it was pretty much the same game and the same way of playing the only difference was how you controlled it so I think people that can't do as much coordination for instance would benefit.
- Yes I think it would be great because it wasn't as tiring as the first one and I basically only used two buttons on the mouse.
- Yes, because it basically does all the same things its just that you can do it in a simpler way without having to tire yourself out as much.
- Yes as the controls are easier to use and don't require as much coordination or button presses

Did the accessible version change the gameplay experience in any way (positively or negatively)?

- It did take off a bit of a mental load because I didn't have to coordinate so much, but at the same time at points it did feel almost like it was playing for me a little bit.
- I think I enjoyed the standard version a bit more purely because I'm used to those controls and it felt a bit more intuitive, and I felt a bit more in control of what I was doing.
- it did feel a bit less challenging because there was less to do so the earlier phases didn't feel as different from each other as they did in version 1 (the difficulty didn't seem to escalate as quickly) but later phases were still challenging and rewarding. overall i found the game itself more enjoyable because i could focus on the gameplay without the added stress of my wrist getting tired or struggling to keep up purely on the physical side of the controls
- Yes it made it a bit more manageable for me and so I had more fun actually eliminating enemies and trying to survive.
- The only downside I felt to the accessible version was I felt more relaxed and less involved in the game, at times it was a little bit like autopilot whereas in the first one because I had to coordinate so much I was more engaged.
- I did feel less pressured and it wasn't as intense and chaotic for me compared to the first one, I think because I didn't have to do as much I was more relaxed.
- I found it more enjoyable because I wasn't getting as annoyed and I could focus more on killing enemies and moving around instead of messing around with my fingers and hands.
- It was less demanding for me which was nice because I could focus more on what was going on in the game instead of on the keyboard and mouse but I enjoyed the movement more on the first one I felt like I had more control and like I could slip through and around enemies better, the clicking to move didnt give me room to manoeuvre around as much

Figure I.4: Interview Responses

Appendix I: Questionnaire and Interview Results

How often do you play games?
8 responses

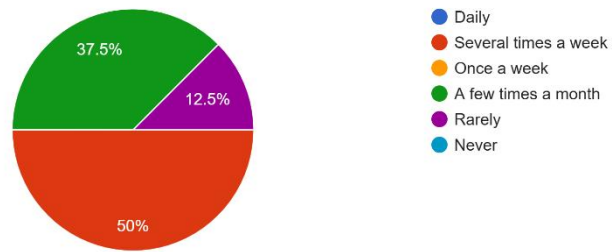


Figure I.5: Gaming Experience Responses

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